hole shrinks end up? Theoretical physicists are debating the question, and no one has a completely clear answer.

All of this, I believe, indicates that in order to grasp the basic grammar of the world, we need to merge three basic ingredients, not just two: not just general relativity and quantum mechanics, but also the theory of heat, that is, statistical mechanics and thermodynamics, which we can also describe as information theory. But the thermodynamics of general relativity, that is to say, the statistical mechanics of quanta of space, is as yet only in its first infancy. Everything is still confused, and there is a very great deal which remains to be understood.

All of this brings us to the last idea I describe in this book: thermal time.

Thermal time

The problem at the root of the idea of thermal time is simple. In Chapter 7, I showed that it is not necessary to use the notion of time to describe physics. It is better to forget time altogether. Time plays no role at the fundamental level of physics. Once we have understood this, it is easier to write the equations of quantum gravity.

There are many everyday notions which no longer have any role in the fundamental equations of the universe; for example, the notions of ‘up’ and ‘down’, or ‘hot’ and ‘cold’, so it is not particularly strange that shared quotidian notions disappear from fundamental physics. However, once we have accepted this idea, we obviously open up a second problem. How can we recover the notions of our everyday experience? How do they emerge, in our specific context?

For example, the notions of ‘up’ and ‘down’ don’t enter into Newton’s equations, but we know what they mean in a schema without absolute up and down. ‘Up’ and ‘down’ are meaningful near a large mass, like a planet. ‘Down’ indicates the direction towards which the large, near mass exerts gravitational pull; ‘up’ indicates the opposite direction. The same goes for ‘hot’ and ‘cold’: there are no ‘hot’ or ‘cold’ things at a microscopic level but, when we put together a large number of microscopic constituents and describe them in terms of averages, then the notion of ‘heat’ appears: a hot body is a body where the average speed of single constituents is raised. We are able to understand the meaning of ‘up’ or ‘hot’ in certain situations: the presence of a nearby mass, or the fact that we are dealing only with average values of many molecules, and so on.

Something similar must apply to ‘time’. If the notion of time has no role to play at an elementary level, it certainly plays a significant role in our lives, just as ‘up’ and ‘hot’ do. What does ‘the passage of time’ mean, if time plays no part in the fundamental description of the world?